

Food Insecurity as a Trigger of Migration Intentions: Evidence from a Pandemic Shock in Five Latin American Countries

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Abstract

Poverty and hunger are well-established drivers of emigration, but the literature largely treats food insecurity as a standing condition rather than something that moves. This paper asks whether a sudden, economy-wide deterioration in food security shifts migration intentions, and whether it does so differently from chronic deprivation. Using the second wave of the World Bank's Phase 2 High Frequency Phone Surveys for Guatemala, El Salvador, Honduras, Nicaragua and the Dominican Republic ($n = 5,565$), we exploit a survey design that records food insecurity both at interview and retrospectively for the pre-pandemic period, allowing current and structural food insecurity to enter the same specification. Severe current food insecurity is associated with an 11.9 percentage point increase in the probability of intending to emigrate (95% CI: 6.9–16.9), against 4.6 points for pre-pandemic food insecurity. When the outcome is instead whether the intention *rose* during the pandemic, current food insecurity remains significant (5.5 points) while the pre-pandemic measure is indistinguishable from zero. Classifying households by their food security transition sharpens the result: the pandemic-era rise in intentions is concentrated among the *newly* food insecure (7.8 points), while the chronically food insecure show no differential rise. Intentions rose across the wealth distribution but most sharply in the top quintile and among the tertiary educated, groups with little direct exposure to hunger, which is consistent with intentions responding to perceived country conditions rather than to private deprivation alone. The results suggest that food security shocks do not merely intensify existing migration pressure but may alter who considers leaving, with implications for the design of shock-responsive social protection in migrant-sending countries.

Keywords: Food insecurity, Migration intentions, COVID-19, Economic shocks, Latin America, High Frequency Phone Surveys

JEL: I15, J61, I31, O15

1. Introduction

Poverty and food insecurity are among the most consistently documented correlates of emigration from low- and middle-income countries (Obi et al., 2020; FAO, 2018; Sadiddin et al., 2019). Almost all of that evidence, however, describes food insecurity as a state rather than as a change. Households are observed as food insecure or not at a

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point in time, and their migration intentions are read against that classification. What happens when food security deteriorates sharply and simultaneously across an entire population is much less well understood, even though shocks of exactly this kind (price spikes, conflict, drought, epidemics) are the situations in which policymakers most need guidance (Lindstrom et al., 2022; Abay et al., 2023; Fujii, 2013).

The distinction is important for policy interventions, too. If migration intentions track the standing level of deprivation, the relevant policy lever is the cumulative generosity of food and income support, and the households to reach are the persistently poor. If instead intentions respond to deterioration, to the household's position relative to where it was, then the timing of assistance matters more than its total volume, and the population at risk includes households that were not poor before the shock and will not appear in any standing beneficiary register. These are different programmes with different budgets, and the existing evidence cannot adjudicate between them, because the two margins are almost never observed separately.

The question is long-standing and present. A decade of work on cash transfers has established that they raise food consumption and reduce food insecurity in low- and middle-income settings (Bastagli et al., 2019), and the debate on the role of the state has highlighted in recent years the urgency to build resilience to expand social protection quickly when a shock arrives (Oxford Policy Management, 2018). Whether that architecture is worth its cost depends in part on which margin of deprivation actually moves household behaviour. If only chronic deprivation matters, a standing programme suffices; if deterioration matters independently, a system that cannot scale within months will miss the households whose behaviour changes.

The COVID-19 pandemic produced a widespread shock at a scale that makes it analytically useful. In Latin America, the share of households reporting having run out of food rose from roughly 17% before the pandemic to almost 32% at its height, and was still near 24% at the end of 2021 (Figure 1). Over the same period the share of Latin Americans declaring an intention to migrate rose from 28% in 2018 to 38% in 2021. Encounters at the southern border of the United States increased 27% between 2021 and 2022, and crossings at the Panama–Colombia border rose 86% (Customs and Border Protection, 2023; International Organization for Migration, 2023). Several authors called at the time for research on precisely this conjunction, hypothesising that the rise in food insecurity would drive a subsequent wave of migration (Smith and Wesselbaum, 2020; Husain et al., 2020).

This paper takes up that question. The analytical difficulty in doing so is that current and chronic food insecurity are ordinarily collinear: households that are hungry today are usually the households that were hungry before, and a single cross-section cannot tell the two apart. We address this using a feature of the World Bank's COVID-19 High Frequency Phone Surveys (HFPS), which asked respondents about food insecurity both at the time of interview and retrospectively about the period before the pandemic. Both measures therefore enter the same regression, and their coefficients can be compared directly. The contrast between them is the empirical core of the paper. The same design permits a sharper test: because the running-out-of-food item is observed for both periods, households can be sorted into mutually exclusive groups: never, newly, chronically, and no longer food insecure. And the question becomes whether, conditional on being food insecure today, it matters that the condition is new.

We use the second wave of Phase 2 of the HFPS for Guatemala, El Salvador, Honduras, Nicaragua and the Dominican Republic ($n = 5,565$), five countries in which emigration is a central economic fact: remittances range from 9% to roughly a quarter of GDP, and emigrant stocks from 9% to 28% of the resident population. Food insecurity was neither rare nor new in the region when COVID-19 arrived. What the pandemic changed was how quickly it spread up the income distribution, and that is the variation the paper exploits. The five countries are also the ones for which the survey's migration and remittance rosters were updated, which is what makes the exercise possible at all; they are, conveniently, among the largest contributors to migration flows to the United States and among the most remittance-dependent economies in the hemisphere, so the sample is not an incidental one. A design caveat follows from the instrument: phone surveys under-cover households without mobile access, who are plausibly both poorer and more food insecure, so the association we report is more likely to be understate than to overstate the relationship in the underlying population.¹

Three findings follow. First, severe current food insecurity is associated with an 11.9 percentage point increase in the probability of intending to emigrate (95% CI: 6.9–16.9), against 4.6 points for recalled pre-pandemic food insecurity: roughly two and a half times as large, against a sample mean of 39%. Second, when the outcome is instead whether the intention *rose* during the pandemic, the two measures separate: current food insecurity remains significant at 5.5 points while the pre-pandemic measure is indistinguishable from zero. The transition classification states the same result without relying on the recalled measure entering linearly: the pandemic-era rise in intentions is concentrated entirely among the newly food insecure (7.8 points, $p < 0.001$), while the chronically food insecure are statistically indistinguishable from households food insecure in neither period. What moves intentions is the deterioration, not the condition. Third, and against the usual gradient, intentions rose most among wealthier and better-educated respondents: 61% of the tertiary educated with a migration intention report that the pandemic strengthened it, against 34% of those with primary schooling or less, and the education gradient survives the full set of controls.

These patterns are not the result of functional form or of any single country. Probit average partial effects are nearly identical to the linear probability estimates, and re-estimating the main specification excluding one country at a time leaves the severe food insecurity coefficient on the intention level between 10.3 and 14.3 points, always significant at the 1% level.

The paper makes three contributions. It separates the association of migration intentions with current versus structural food insecurity, a distinction the literature has largely neglected and which turns out to carry the result. It exploits a second recency outcome (whether the respondent reports that the intention increased *because of* the pandemic) that isolates the shock-driven component of intentions from their standing level. And it documents a compositional shift: there was a movement in intention in households least exposed to hunger directly, which is consistent with intentions responding to perceived country trajectory rather than to private deprivation alone, and which implies that a food security shock may have strong spillover effects.

¹The HFPS took special attention in making the sample nationally comparable, see the section on data.

Three limitations should be stated at the outset. The data are cross-sectional, so the estimates are conditional associations rather than causal effects; Section 4 sets out what we can and cannot rule out. The pre-pandemic measure is recalled, and recall error is likely correlated with current conditions in a way that attenuates its coefficient; we treat the magnitude of the contrast as an upper bound and show in Section 5 that attenuation cannot explain the sign reversal on the second outcome. And intentions are not migration: the gap between the two is systematic rather than random, and is itself a function of wealth (Smith and Floro, 2020). Section 5.6 treats this directly.

The remainder of the paper is organised as follows. Section 2 sets out the conceptual framework, reviews the evidence linking food insecurity to migration, and describes the five sending countries. Section 3 describes the surveys and variables, Section 4 the empirical approach. A results and discussion Section describes mechanisms and heterogeneity, and Section 6 draws policy implications.

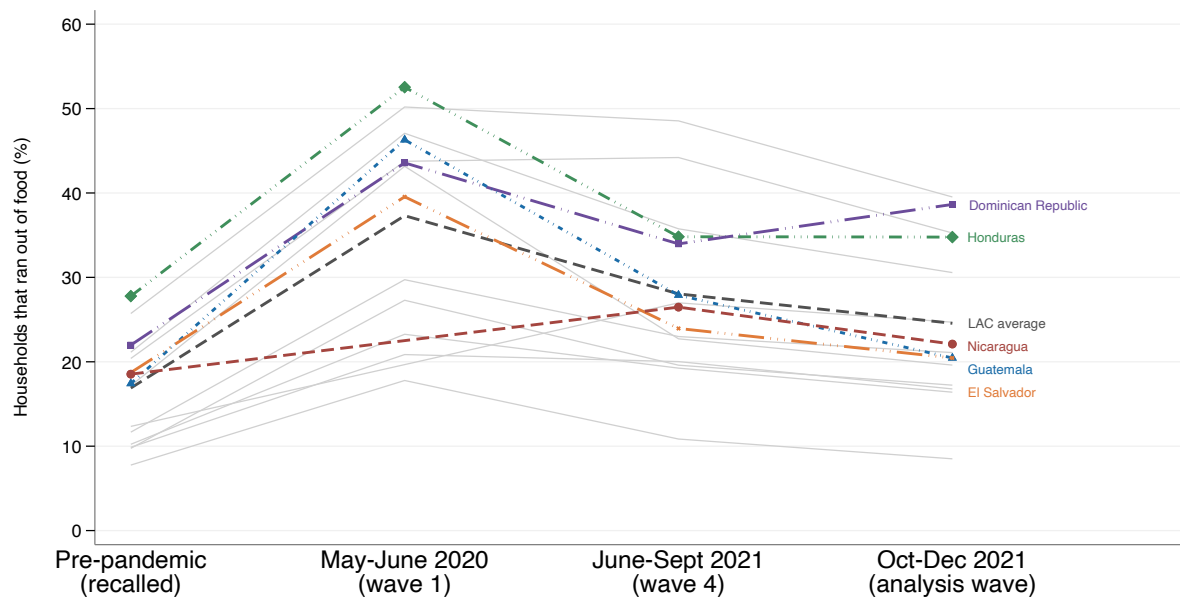


Figure 1: Food insecurity before and during the COVID-19 pandemic, five sending countries and the wider region

Notes: Share of households reporting having run out of food in the previous 30 days for lack of money. Coloured lines are the five countries studied; grey lines are the remaining Latin American and Caribbean countries in the HFPS. The dashed dark line is the unweighted mean across countries observed in all four periods ($n = 13$). The pre-pandemic value is recalled at the first interview. Nicaragua was not fielded in waves 1–3, so its line is interpolated between the periods in which it appears. *Source:* Authors' calculations from the World Bank HFPS, Phases 1 and 2.

2. Conceptual framework and setting

2.1. Food insecurity in the migration decision

We start the discussion thinking in terms of the new economics of labour migration, by which migration is a household strategy for smoothing consumption and diversifying risk (Stark and Bloom, 1985; Stark and Lucas, 1988;

Hoddinott, 1994). Where insurance and credit markets are thin, sending a member abroad substitutes for the contracts households cannot buy, and the decision weighs present needs against the monetary and social costs of leaving (De Jong and Gardner, 2014; Clark and Lisowski, 2019). Food insecurity enters this calculus in two ways that pull in opposite directions. It raises the return to holding income claims outside the local economy, which strengthens the incentive to send a migrant. But it also depletes the resources that departure requires, which weakens the capacity to act (Dustmann and Okatenko, 2014). The aspirations–capabilities framework makes this tension explicit: the wish to migrate and the ability to realise it are analytically distinct, and deprivation can raise the first while lowering the second, producing involuntary immobility rather than movement (Carling, 2002; de Haas, 2021; Carling and Schewel, 2018). Because our outcome is an intention rather than a departure, it sits on the aspiration side of that distinction, where the first mechanism should dominate. This is also why the sign of the association we estimate is informative but its magnitude is not a migration elasticity: a household can revise its intention upward at no cost, and the resource depletion that accompanies food insecurity bites later, at the point of departure.

Framing food insecurity as a shock rather than a state has a further implication that the literature has not exploited. If migration decisions are forward-looking, what should enter them is not the household’s history but its expectation of the future, and a household’s current position is a better signal of that expectation than its past is. A household that was food insecure before the pandemic and is not now has learned that its situation can improve; a household newly food insecure has learned that its situation can deteriorate. The two may be observationally identical in a single-period measure of deprivation and yet hold opposite beliefs about what comes next. This is the mechanism our design is built to detect, and it generates a testable prediction: current food insecurity should dominate recalled food insecurity in explaining intentions, and should do so most clearly for the component of intentions attributable to the shock.

Relative position matters alongside absolute deprivation. In the relative-deprivation tradition, households compare themselves with a local reference group, and it is the gap rather than the level that generates the incentive to move (Stark and Taylor, 1991). Recent evidence carries this into the food security domain: at the country level, food insecurity *inequality* predicts emigration beyond the effect of its level (Smith and Wesselbaum, 2022), and inequality more generally predicts potential emigration across the Gallup World Poll (Nikolova, 2023). This matters for interpreting our heterogeneity results, because it implies that households need not be food insecure themselves for a visible deterioration around them to shift their intentions.

2.2. *From condition to shock*

The empirical literature linking food insecurity to migration is by now substantial and consistent in sign. Food insecurity predicts migration intentions across low- and middle-income countries and does so monotonically in severity (Smith and Floro, 2020; Migali and Scipioni, 2018; Sadiddin et al., 2019); at the country level it predicts realised flows (Smith and Wesselbaum, 2022); and in specific settings the mechanism has been traced through agricultural livelihoods and the gendered reorganisation of household labour that migration itself sets in motion (Ghimire et al., 2023; Azzarri and Zezza, 2011). A parallel literature on environmental drivers reaches the same conclusion from the

other direction, showing that weather and temperature shocks operating through agricultural income raise emigration (Cattaneo and Peri, 2016; Missirian and Schlenker, 2017; Meierrieks, 2021).

What this body of work has not separated is the standing condition from its deterioration. The environmental studies come closest, since a drought is by construction a change, but they identify the shock in rainfall or temperature rather than in food security itself, and they cannot say whether the households responding are those newly deprived or those already at the margin. Studies that measure food insecurity directly almost always do so at a single point in time, so the coefficient on food insecurity mixes the effect of being poor with the effect of having become poorer. Separating the two requires observing the same household in two states, which panel data would deliver and which cross-sections generally do not.

The COVID-19 pandemic generated the requisite variation. Food security deteriorated sharply and broadly across low- and middle-income countries (Josephson et al., 2021; Amare et al., 2021; Adjognon et al., 2021; Rudin-Rush et al., 2022; Picchioni et al., 2022), including in Latin America (Giacoman et al., 2021), and it did so through income and mobility channels that reached households well outside the pre-pandemic poor (Ouoba and Sawadogo, 2022; Committee on World Food Security, 2021). The phone-survey infrastructure assembled to track those effects in near real time has since been used to study welfare under conflict and other acute shocks (Abay et al., 2023; Kaila and Azad, 2023). We use the same infrastructure, and the retrospective food security question it carries, to ask a question that infrastructure has not yet been used to answer.

2.3. Intentions as an outcome

Because we observe intentions rather than departures, how intentions should be read requires a word. They are informative but not equivalent. Across country pairs, changes in stated emigration plans track changes in realised flows with an elasticity of roughly 0.75 (Tjaden et al., 2019), and potential migration predicts actual migration most strongly where networks at destination lower the cost of moving (Docquier et al., 2014; Manchin and Orazbayev, 2018). Conversion is nonetheless selective: realising an intention requires cash for travel, documents and settlement, so wealth constraints bind at the bottom of the distribution (Dustmann and Okatenko, 2014). Both points matter here. Intentions are a meaningful leading indicator of pressure, which is what makes them useful to policy; and their translation into departures depends on who forms them, which is why the compositional result in Section 5 carries weight beyond the coefficient itself.

2.4. The five sending countries

The analysis covers Guatemala, El Salvador, Honduras, Nicaragua and the Dominican Republic. The four Central American countries are neighbours with a closely shared history (after independence they briefly formed a single state, and today the CA-4 treaty allows free movement of people across their borders) while the Dominican Republic has followed a diverging path, building a large tourism sector and reaching the highest income level of the five. All

are middle-income: Nicaragua and Honduras in the lower band, with GDP per capita of \$2,600 and \$3,300, the other three in the upper band (Table 1).

What unites them for our purposes is the weight of migration in their economies. Remittances range from 9% of GDP in the Dominican Republic to roughly a quarter in El Salvador, Honduras and Nicaragua, placing them, with Haiti, at the top of the Latin American ranking of remittance dependence (Knomad, 2020, 2022a). Emigrant stocks are correspondingly large, from 9% of the resident population in Guatemala to 28% in El Salvador, and the United States is the dominant destination for all five, Nicaragua partially excepted owing to its historical corridor to Costa Rica. The Dominican Republic is distinctive in being both a sender and a receiver, with a sizeable Haitian labour force. Remittance inflows proved resilient through the pandemic, which matters for interpretation: the shock we study operated on domestic food security rather than on the transfers that partly insure against it (Knomad, 2022b; Mora-Rivera and Van Gasteren, 2021).

The five also share the conditions that make food insecurity a plausible migration push factor. Poverty at the \$3.65 line reaches roughly a quarter of the population in Guatemala and Honduras, and FAO estimates put moderate or severe food insecurity between 41% and 53% of the population before the pandemic, rising in every country except the Dominican Republic by 2021–23 (Table 1). All five operate conditional cash transfer programmes and receive sizeable external food assistance, but coverage is concentrated on the structurally poor—which is precisely the group our results suggest was not the one whose migration intentions moved. Emergency assistance to the region expanded during the pandemic, but it was delivered largely through humanitarian rather than national transfer channels and was targeted geographically on the areas of chronic deprivation identified before the shock (Husain et al., 2020). The mismatch between where assistance is directed and where intentions moved is a theme we return to in Section 6.

Table 1: Economic and migration indicators, selected countries

	Guatemala	El Salvador	Honduras	Nicaragua	Dominican Rep.
GDP per capita (USD)	5,739	5,550	3,285	2,599	11,825
Population (millions)	16.9	6.5	9.9	6.6	11.9
Remittances (% of GDP)	19	24	26	26	9
Poverty rate, \$3.65 (%)	26	8	27	14	4
Food insecurity FAO 2017–19 (%)	45.2	42.2	40.9	n.a.	52.6
Food insecurity FAO 2021–23 (%)	59.8	46.9	56.0	n.a.	46.1
Emigrant stock (millions)	1.5	1.8	1.4	0.9	1.9
Emigrant stock (% of population)	9	28	14	14	16

Notes: GDP per capita, population and poverty rates are for the latest available year from the World Development Indicators. Food insecurity is the FAO prevalence of moderate or severe food insecurity (three-year averages). Emigrant stocks from IOM; the percentage divides the emigrant stock by the resident population.

3. Data

3.1. The COVID-19 High Frequency Phone Surveys

The analysis uses the 2020–2021 High Frequency Phone Surveys collected by the World Bank (2022) and partners. Launched at the onset of the pandemic to track its effects in near real time, the programme produced more than 400 rounds across more than 100 countries, covering social and economic conditions, wealth and income, exposure to shocks, government assistance and labour, with specialised modules on remittances and migration. The design traded the depth of a face-to-face survey for speed and repetition, and the data have since supported a substantial literature on welfare under acute shocks (Josephson et al., 2021; Rudin-Rush et al., 2022; Abay et al., 2023; Kaila and Azad, 2023).

The Latin American and Caribbean rounds were largely harmonised at deployment (a common core questionnaire, a common reference period and a common fieldwork protocol) which is what makes pooling across countries defensible here. Five waves were fielded over 2020–2021: three immediately after the onset of the pandemic (Phase 1) and two during 2021 (Phase 2). The Phase 2 rounds were fielded in tight windows: the first across the region in May and June 2021, with Haiti (July–August) and Brazil (August–September) the exceptions, and the second between October and December 2021, extending into January 2022 for Peru. In all five countries studied here the second round was fielded between October and December 2021.

The windows matter for two reasons: they are narrow enough that pooling the five cross-sections does not conflate distinct phases of the pandemic, and they fix the recall period for the retrospective food security item at roughly eighteen months.

The survey used a dual-frame Random Digit Dialing (RDD) design that sampled both landline and mobile phone numbers, ensuring coverage of all active telephone numbers under each country’s national numbering plan. Because the survey is representative at the national level, sampling weights were constructed for both households and individuals by accounting for selection probabilities, non-response, attrition in later waves, and calibration to official population statistics adjusted for telephone coverage.

We use the second wave of Phase 2 and restrict attention to Guatemala, El Salvador, Honduras, Nicaragua and the Dominican Republic. The restriction is driven by data availability: the migration and remittance rosters were updated for these five countries only. This does not drive down the credibility of the estimates, since these are among the largest contributors to migration flows to the United States and among the largest remittance recipients relative to GDP (Knomad, 2020, 2022a). The five nationally representative surveys yield 5,565 observations. Dropping observations with missing information leaves estimation samples of 5,264 to 5,303 depending on the outcome, a loss of 4.7% to 5.4%.²

²That loss falls on the wealth-index components rather than on the food security or migration items, so the households dropped are not disproportionately those the analysis is about.

One feature of the design deserves emphasis rather than a footnote. Telephone surveys exclude households without mobile (or landline) access and under-represent those with intermittent access, and coverage failures of this kind are not random: they correlate with income, rurality and education (Lipps, 2016; Weeks et al., 1983; Carr and Worth, 2001). The households outside the frame are plausibly poorer and more food insecure than those inside it, with two consequences. Our prevalences understate food insecurity in the underlying population, which is why we benchmark them rather than present them as national rates. And because the excluded households sit where the association between food insecurity and migration intentions is strongest, our coefficients are more likely attenuated than inflated. Telephone and face-to-face administration of factual items of this kind generally agree, which limits but does not remove the concern (Scherpenzeel et al., 2020).

3.2. Outcome variables

The migration roster first asks whether the respondent has any desire to migrate within the following three years (*Desire to migrate*). Conditional on a positive answer, it asks whether that desire has increased, decreased or stayed the same as a result of the pandemic. We code *Increased desire to migrate* as one where the respondent reports an increase.

Using both outcomes is deliberate. The first captures the level of migration intention, which reflects the accumulated influence of structural conditions; the second captures movement in that intention attributable to the shock, and is closer to the object of policy interest. Two features of the wording bear on interpretation. The second item asks the respondent to attribute a change to the pandemic, which is a subjective causal claim rather than an intention differenced across two measurements, we therefore read it as a report of perceived shock-driven movement. And because it is asked only of those answering yes to the first, it describes intensification among people already contemplating migration rather than entry into the pool, a conditioning that matters in Section 5, where the conditioning set differs in size across groups. Section 5 benchmarks our intention levels against the AmericasBarometer and Latinobarómetro; question wording differs across those instruments, so the comparison bears on order of magnitude and country ranking, not on point estimates.

3.3. Food insecurity

Food insecurity is measured through three questions on food availability and quality, plus a fourth on pre-pandemic experience, with a three-month reference period as is standard in the region. The items are: (i) skipping a meal for lack of resources, asked both currently and retrospectively for the pre-pandemic period; (ii) going without food for a full day for lack of resources; and (iii) not eating healthy food for lack of resources.

We construct two indicators. *Severe food insecurity* equals one where the respondent reports all three conditions, including going without food for a full day. *Moderate food insecurity* equals one where the respondent reports at least one of the two less extreme conditions. By construction the severely food insecure also satisfy the moderate definition, so the two enter separate specifications rather than the same one.

These thresholds follow FAO practice in spirit but are not identical to the Food Insecurity Experience Scale, which uses eight items and a Rasch-model adjustment to place countries on a common latent scale (Cafiero et al., 2018; Ballard et al., 2014; Nord et al., 2016). A three-item battery cannot support that adjustment, so our indicators are ordinal within the sample but not calibrated across countries; department fixed effects absorb country-level differences in how the items are understood. As a plausibility check, our moderate food insecurity rate of 35% sits below the FAO prevalence of moderate or severe food insecurity for these countries (41–53% before the pandemic, Table 1), the direction both the narrower item set and the telephone frame would predict.

Because both current and recalled pre-pandemic food insecurity are observed, the two can be included jointly. This is the feature of the data the paper exploits, and its price is worth stating here rather than only in the discussion: the pre-pandemic measure is retrospective, given roughly eighteen months after the period it describes. Recall of episodic hardship decays and is likely coloured by present circumstances, so the recalled measure is noisier and its coefficient attenuated. Section 5 sets out why this cannot account for the paper’s main result.

The running-out-of-food item is the only one asked for both periods, so it also allows a mutually exclusive classification into four groups: never food insecure on this item (65.6% of the sample), *newly* food insecure (14.0%), *chronically* food insecure (12.0%), and no longer food insecure (8.4%). Section 5 uses this directly. Its appeal is that it does not require the recalled measure to enter linearly alongside the current one; it asks only that respondents can order their own before-and-after status on a single concrete item.

3.4. Controls

We include demographic controls (gender, age, marital status, urban residence) and education (primary, secondary, tertiary). Household wealth and coping behaviour enter through: (i) a country-specific wealth index constructed by principal component analysis over ownership of or access to a dishwasher, PC or tablet, internet, motorcycle, potable water at home, and an overcrowding indicator; (ii) receipt of any government assistance during the pandemic; and (iii) use of savings or non-payment of debt as a coping strategy. The index is estimated separately by country because the same asset signals different positions in different distributions; the first component explains 24 to 42% of the variance, depending on country.

Government assistance, coping behaviour and remittance receipt are included because each mediates the relationship between a shock and its consumption consequences; omitting them would load their effect onto food insecurity, and transfers from abroad are the region’s principal private insurance mechanism (Mora-Rivera and Van Gameren, 2021). We also include an indicator for increased time spent on household chores: pandemic-era shifts in household time allocation were large and strongly gendered, and could plausibly bear on who considers leaving. About 25% of respondents report spending more time on chores.

Table 2: Descriptive statistics

	All Sample	GTM	SLV	HND	NIC	DOM
<i>Migration Intention</i>						
Migration intention	0.39	0.33	0.34	0.46	0.34	0.46
Migration intention up since pandemic	0.20	0.15	0.16	0.28	0.18	0.23
<i>Food Insecurity - FI *</i>						
Severe FI	0.07	0.03	0.03	0.11	0.06	0.13
Moderate FI	0.35	0.25	0.27	0.47	0.34	0.46
FI Pre-Pandemic	0.20	0.13	0.18	0.30	0.18	0.24
<i>Demographics</i>						
Married	0.58	0.57	0.58	0.60	0.61	0.55
Age	38.15	36.90	40.77	36.56	37.20	39.74
Woman	0.52	0.51	0.51	0.57	0.48	0.54
Urban area	0.62	0.60	0.62	0.55	0.65	0.68
Municipal Migration Rate **	0.10	0.05	0.11	0.16	0.11	0.10
Up to Primary	0.47	0.46	0.50	0.48	0.41	0.51
Secondary Ed.	0.40	0.44	0.31	0.43	0.39	0.39
Tertiary Ed.	0.13	0.10	0.19	0.09	0.20	0.11
<i>Household Finances</i>						
Wealth Index ***	0.02	0.03	0.01	0.05	0.01	0.00
Income down	0.29	0.26	0.25	0.28	0.30	0.35
Difficulty Making Ends	0.51	0.44	0.52	0.56	0.52	0.56
Remittances before the pandemic	0.22	0.11	0.22	0.27	0.27	0.28
<i>Gender and Chores</i>						
Increased time on household chores	0.25	0.24	0.20	0.25	0.24	0.30
Chores time up x women	0.15	0.15	0.12	0.16	0.13	0.17
<i>Coping</i>						
Any kind of Gov Help	0.40	0.32	0.81	0.29	0.18	0.47
Remittances during the pandemic	0.25	0.14	0.25	0.30	0.26	0.32
Coping: Savings using or didn't pay debts	0.62	0.53	0.56	0.70	0.60	0.74
<i>N</i>	5264	1466	791	922	842	1243

Notes: Unweighted means. Food insecurity is severe if the respondent reports all three food insecurity conditions, including being unable to eat for a whole day. Moderate food insecurity indicates a non-nutritious diet or having run out of food. The pre-pandemic question is asked retrospectively. Municipal migration rates are constructed from census data, except for Honduras, where comparability problems require household survey data. The wealth index is a country-specific principal component index over ownership of or access to a dishwasher, PC/tablet, internet, motorcycle, potable water at home, and overcrowding (more than two adult equivalents per room).

3.5. Auxiliary data

Census-based migration rates. Migration networks shape both subsequent migration and remittance receipt (Mora-Rivera and Van Gasteren, 2021; Acosta et al., 2006; Adams and Cuecuecha, 2010) and condition whether an intention can be realised (Manchin and Orazbayev, 2018). We therefore merge municipal migration rates from census sources, using national statistical offices and IPUMS (Minnesota Population Center, 2020). Table 3 reports the level of merge: most countries merge at municipality level, while data limitations restrict the Dominican Republic to province level. This yields between 32 and 246 unique geographic units against surveys averaging roughly 1,000 observations per country, so the merged variable is measured with more error there than elsewhere. Census rounds also differ in vintage across the five, so the variable captures the durable geography of migration rather than its recent evolution, which is the margin we want it to control for.

Table 3: Share of households with a migrant member

Country	Level	Year	Total Regions	Mean	SD	Max	Min
Guatemala	Municipality	2018	246	5.2%	0.044	37.1%	0.4%
El Salvador	Municipality	2007	154	10.7%	0.054	40.2%	2.7%
Honduras	Municipality	2013	196	4.9%	0.024	21.9%	0.2%
Nicaragua	Municipality	2005	122	10.4%	0.042	21.5%	0.8%
Dominican Republic	Province	2010	32	9.5%	0.017	14.0%	4.3%

Notes: Census data for Guatemala, Honduras and Nicaragua via the REDATAM web interface; other countries from IPUMS harmonised census samples. *Level* indicates the geographic level of the merge to survey data. *Total regions* is the number of unique units at that level. Statistics use survey weights.

AmericasBarometer We use it to benchmark our migration-intention levels against independent regional series.

4. Empirical approach

Figure 2 sets out the assumed structure. Illness and lockdowns impose direct and indirect costs; households respond with short-term coping strategies; where these are insufficient, financial distress and food insecurity follow; and that deterioration may raise migration intentions. The diagram is a statement of the channel we have in mind, not a claim that we identify each arrow separately.

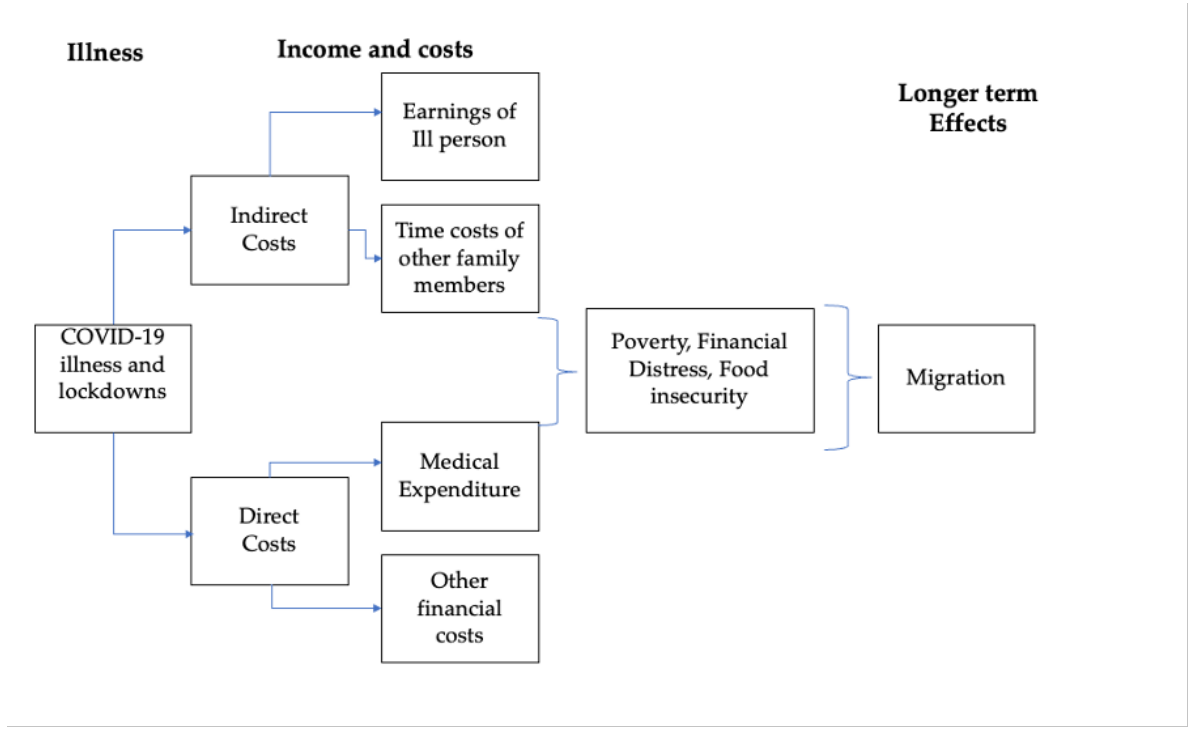


Figure 2: Assumed channels from health shock to migration intention

4.1. Specification

We estimate a linear probability model:

$$\text{Migrate}_i = \beta_0 + \beta_1 \text{FI}_i + \beta_2 \text{FI}_i^{\text{pre}} + \mathbf{X}_i' \boldsymbol{\gamma} + \Psi_d + \epsilon_i \quad (1)$$

where Migrate_i is an indicator for expressing a desire to migrate or, in the alternative specification, for that desire having increased; FI_i is current food insecurity and FI_i^{pre} its recalled pre-pandemic counterpart; $\mathbf{X}_i$ collects the controls described in Section 3; and Ψ_d are department (first-level administrative unit) fixed effects, which are below country effects.

The parameter of interest is not β_1 or β_2 alone but the contrast $\beta_1 - \beta_2$: whether current food insecurity is associated with migration intentions over and above the household's standing condition. Because both regressors are measured for the same household in the same interview, that contrast differences out any household-level factor that shifts both reports equally, a respondent's general propensity to report hardship, for instance, in a way that neither coefficient does on its own.

We use a linear probability model for interpretability and because the contrast of two coefficients has no natural analogue in a nonlinear index model; probit average partial effects, reported in Appendix A, are nearly identical. A complementary specification replaces the two indicators with the mutually exclusive transition categories defined in Section 3, newly, chronically, and no longer food insecure, with the never food insecure as the omitted group, keeping

controls and fixed effects unchanged. This asks the sharper question: conditional on being food insecure today, does it matter that the condition is new?

4.2. Inference

Standard errors for the regressions presented in the main text are heteroskedasticity-robust. In Appendix B i report different approaches: clustering with 103 departments, there is almost no movement of the SE; on the other hand, clustering at the 5 countries seems extremely conservative, but results are also reported for reference.

Estimates are unweighted. Survey weights are designed to recover population means, and using them to estimate a conditional relationship trades efficiency for a robustness that matters only if the model is misspecified in a way that correlates with the weights (Solon et al., 2015). Since the weights here are built from the same demographics that enter $\mathbf{X}_i$ as controls, that correlation should be limited. We report weighted estimates in Appendix E; the coefficients of interest are unchanged in substance.

We report four principal specifications, two food insecurity measures crossed with two outcomes, plus the transition variant, so some attention to multiplicity is warranted. The result the paper rests on is a sign reversal rather than a marginal rejection, and would survive any reasonable correction; for completeness Appendix C reports Romano–Wolf stepdown p -values across the family (Romano and Wolf, 2005).

4.3. What the estimates identify

The data are a single cross-section, so these are conditional associations. Two threats deserve stating plainly.

First, reverse causality. A household already planning to migrate may run down assets to finance departure and so report food insecurity. This would bias β_1 upward. It is harder to see how it biases the *contrast*, since a household financing a planned departure has no reason to misreport its pre-pandemic status differentially.

Second, unobserved local deterioration (a closed factory, a failed harvest) could drive both food insecurity and migration intentions. Department fixed effects absorb shocks common to a department, including national ones, but not variation at finer geographic levels. Here the transition specification helps: a local shock that pushed a community into food insecurity would move the newly and the chronically food insecure alike, whereas we find movement only among the newly food insecure.

Neither argument delivers exogeneity, and we do not claim it. What they establish is that the contrast is more robust to the obvious confounders than either level is, which is why we treat it as the paper’s finding and the levels as descriptive. Appendix D reports an Oster (2019) bound on selection on unobservables for the main specification.

5. Results and discussion

5.1. Descriptive patterns

Table 2 reports summary statistics. About 7% of respondents report severe food insecurity and 35% moderate food insecurity, while roughly one in five reports having been food insecure before the pandemic. Honduras and the

Dominican Republic show the heaviest burden. On the outcome side, about 39% express a desire to migrate, and 20% of the full sample, roughly half of those with a desire, report that the desire increased during the pandemic. The countries with the highest food insecurity are also those with the highest migration intentions, which is the cross-country counterpart of the household-level association we estimate below.

Respondents average 38 years of age with a roughly balanced gender distribution; most are married, urban, and have completed at most primary education. The economic backdrop is severe: 29% report a fall in income and 51% report difficulty making ends meet. Coping is widespread, 40% received some government assistance and 25% received remittances during the pandemic, while 62% report drawing down savings or failing to service debt. These last figures matter for interpretation. A household that has exhausted its savings and is nonetheless food insecure has already used the instruments that would ordinarily absorb a shock, which is the condition under which the new economics of labour migration predicts that a household turns to migration as the remaining diversification strategy.

Because phone surveys under-cover the poorest households, the levels in Table 2 should be read against an independent series rather than as national rates. The natural comparator is the AmericasBarometer, whose migration item shares our three-year horizon almost verbatim, *do you have any intention of going to live or work in another country in the next three years?*, and which is administered face to face (LAPOP Lab, 2021). Across the five countries its 2021 round puts the intention rate at 43.4%, against 32.1% in the 2018–19 round.³ Our 39% therefore falls between the two rounds and below the contemporaneous benchmark, which is the direction the telephone frame predicts. Two further differences push the same way: the 2021 AmericasBarometer was fielded between March and August 2021, four to six months before our wave, and it interviews individuals in person rather than by phone.

Latinobarómetro is not a usable benchmark for this quantity. Its migration item asks whether the respondent *and their family* have *ever* considered going to live abroad, with no time horizon and at family rather than individual level; it returns 31.2% across the same five countries in 2020, but the number is not measuring the same thing and we do not use it.

5.2. Food insecurity and the level of migration intentions

Table 4 reports the main estimates. Households experiencing higher levels of food insecurity are more likely to express a desire to migrate, after conditioning on demographic and economic characteristics and department fixed effects. We focus on severe food insecurity (columns 3 and 4); moderate food insecurity behaves similarly in sign but is smaller in magnitude, consistent with the monotonicity in severity reported elsewhere (Smith and Floro, 2020; Sadiddin et al., 2019).

Pre-pandemic food insecurity is associated with a 4.6 percentage point increase in the probability of wanting to emigrate. Current food insecurity is associated with an increase of 11.9 points (95% CI: 6.9–16.9), roughly two and a

³Weighted country means, then averaged unweighted across the five: Honduras 54.0%, Nicaragua 51.9%, Dominican Republic 43.5%, Guatemala 36.0%, El Salvador 31.6% in 2021.

half times as large, against a sample mean of 39%. The two coefficients are estimated on the same households from the same regression, so the comparison does not rest on any assumption about comparability across samples.

The estimates use 5,264 observations and explain 15.1 per cent of the variation in the intention level. The controls behave largely as the literature would predict, and three deserve comment. The municipal migration rate enters positively (5.4 points per ten percentage points, $p = 0.03$), consistent with networks lowering the perceived cost of moving and with the network results in Manchín and Orazbayev (2018); it also absorbs geographic variation that would otherwise load onto food insecurity. Remittance receipt is associated with a higher probability of intending to migrate, whether received during the pandemic (8.7 points) or before it (4.8 points), which favours reading remittances as marking an established corridor rather than as insurance that substitutes for departure. Government assistance is indistinguishable from zero, a coefficient we read with caution in any case, since receipt is itself a function of pre-existing poverty and of programme geography rather than of the shock. Age and marital status carry the usual signs, with intentions declining in age.

5.3. *The shock margin*

The second outcome sharpens the contrast. When the dependent variable is whether the desire to migrate *rose* during the pandemic, the coefficient on pre-pandemic food insecurity is -0.011 and statistically indistinguishable from zero, while current food insecurity remains significant at 5.5 points. Structural food insecurity is associated with the standing level of migration intention but not with its movement during the shock; current food insecurity is associated with both.

The magnitudes are economically meaningful rather than merely significant. Against a mean of 39%, an 11.9 point difference is a 30% relative increase in the probability of intending to emigrate; against the 20% of the sample reporting an intensified intention, the 5.5 point coefficient is a 28% relative increase. These are large for a single household characteristic, and of the same order as the gaps the cross-country literature reports between the food secure and the severely food insecure (Smith and Floro, 2020; Migali and Scipioni, 2018). That our estimate is comparable despite a narrower food security instrument and a telephone frame that truncates the bottom of the distribution suggests the association is not an artefact of measurement.

This asymmetry is the paper's central result, and two readings are consistent with it. The first is salience: acute deprivation during an unprecedented disruption may crowd out consideration of past hardship, in line with the general tendency to prioritise immediate threats. The second, and to us more plausible, is informational, and follows the prediction set out in Section 2. Current food insecurity is a better signal of expected future conditions than a household's history is. A household that was food insecure before the pandemic and is not now has evidence that its situation can improve; one that is food insecure now has evidence that it can deteriorate. If intentions are forward-looking, they should track the current signal, which is what we observe.

Measurement error in the recalled variable will attenuate its coefficient and so mechanically widen the gap between the two. This cannot explain the sign reversal on the second outcome, attenuation pushes a coefficient toward zero,

404 not from positive to negative, but it does mean the magnitude of the contrast should be read as an upper bound.

Table 4: Food insecurity and migration intentions

Dependent Variable	(1) Migration intention	(2) Migration intention up since pandemic	(3) Migration intention	(4) Migration intention up since pandemic
<i>Food Insecurity - FI</i>				
Severe FI			0.119*** (0.025)	0.055** (0.021)
Moderate FI	0.091*** (0.016)	0.052*** (0.013)		
FI Pre-Pandemic	0.033+ (0.017)	-0.021 (0.014)	0.046** (0.017)	-0.012 (0.014)
<i>Demographics</i>				
Married	-0.087*** (0.013)	-0.012 (0.011)	-0.087*** (0.013)	-0.012 (0.011)
Age	-0.006*** (0.000)	-0.004*** (0.000)	-0.006*** (0.000)	-0.003*** (0.000)
Woman	-0.125*** (0.015)	-0.039** (0.012)	-0.122*** (0.015)	-0.037** (0.012)
Urban area	0.016 (0.015)	0.014 (0.012)	0.015 (0.015)	0.013 (0.012)
Municipal Migration Rate	0.513* (0.243)	0.241 (0.201)	0.522* (0.244)	0.247 (0.202)
Secondary Ed.	0.003 (0.015)	0.055*** (0.012)	0.000 (0.015)	0.053*** (0.012)
Tertiary Ed.	0.004 (0.022)	0.101*** (0.018)	-0.002 (0.022)	0.097*** (0.018)
<i>Household Finances</i>				
Wealth Index	-0.010+ (0.005)	0.004 (0.004)	-0.013* (0.005)	0.003 (0.004)
Income down	0.047** (0.015)	0.059*** (0.012)	0.053*** (0.015)	0.063*** (0.012)
Difficulty Making Ends	0.047*** (0.014)	0.066*** (0.012)	0.056*** (0.014)	0.072*** (0.011)
Remittances pre-pandemic	0.047* (0.020)	0.029+ (0.016)	0.048* (0.020)	0.030+ (0.016)
<i>Gender and Chores</i>				
Increased time on chores	-0.006 (0.022)	0.131*** (0.019)	-0.005 (0.022)	0.132*** (0.019)
Chores time up and women	0.021 (0.029)	-0.050* (0.024)	0.022 (0.030)	-0.050* (0.024)
<i>Coping</i>				
Any kind of Gov. Help	-0.007 (0.014)	-0.023+ (0.012)	-0.005 (0.014)	-0.021+ (0.012)
Remittances during the pandemic	0.087*** (0.019)	0.056*** (0.016)	0.087*** (0.019)	0.057*** (0.016)
Coping: Savings/Debts	0.131*** (0.014)	0.110*** (0.012)	0.139*** (0.014)	0.116*** (0.012)
Observations	5264	5303	5264	5303

Notes: Robust standard errors in parentheses. Significance: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Each column reports a separate regression. The dependent variable is *migration intention* (columns 1, 3) or *increase in migration intention due to the pandemic* (columns 2, 4). Omitted education category is up to primary. Department (first-level administrative unit) fixed effects included.

5.4. New versus chronic food insecurity

Table 5 replaces the two food insecurity indicators with the mutually exclusive transition categories, using the running-out-of-food item observed for both periods. The level of migration intentions (column 1) responds to being food insecure now, whether the condition is new (9.9 points) or chronic (11.5 points); households that escaped food insecurity retain a smaller and marginally significant elevation (5.4 points), consistent with scarring or with persistent unobservables.

The second outcome separates the three groups cleanly. The pandemic-era *rise* in intentions is concentrated entirely among the newly food insecure (7.8 points, $p < 0.001$); the chronically food insecure are statistically indistinguishable from the never food insecure (2.2 points, n.s.), as are those who escaped it. Households for whom the pandemic *created* food insecurity are thus the ones whose intentions moved, which is the paper's claim in its sharpest form: what shifts migration intentions is the deterioration, not the condition.

This specification is the more credible of the two, for a reason worth making explicit. It does not require the recalled measure to enter linearly alongside the current one, nor that the two be measured with comparable precision. It asks only that respondents can correctly order their own before-and-after status on a single concrete item, a far weaker demand on recall than reporting a level. That the two specifications agree, and that the transition version delivers the sharper separation, is evidence that the contrast is not an artefact of differential measurement error.

It is worth being clear about what the transition result does not say. It does not say that chronic food insecurity is unrelated to migration: column 1 shows the chronically food insecure have the highest intention levels of any group, 11.5 points above the never food insecure. It says their intentions did not *move* during the pandemic, which is what one would expect if those intentions were already formed and already high. The policy implication is not that chronic deprivation can be ignored, but that a shock adds a distinct population to the one standing programmes already serve.

Table 5: Migration intentions by food security transition

Dependent Variable	(1)	(2)
	Migration intention	Migration intention up since pandemic
Newly food insecure	0.099*** (0.021)	0.078*** (0.019)
Chronically food insecure	0.115*** (0.023)	0.022 (0.019)
No longer food insecure	0.054* (0.024)	0.008 (0.020)
Full controls	Yes	Yes
Department fixed effects	Yes	Yes
Observations	5263	5302

Notes: Robust standard errors in parentheses. Significance: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Transition categories are defined on the running-out-of-food item, asked for both the current and the pre-pandemic period: *newly* (insecure now, not before), *chronically* (both periods), and *no longer* food insecure (before, not now). The omitted category is households food insecure in neither period. Full controls as in Table 4.

5.5. Who is newly considering migration

The level and the movement of intentions sit on opposite ends of the social gradient (Figure 3). Intention levels fall with advantage, as the standard gradient predicts: 34–35% of respondents in the top two wealth quintiles express a desire to migrate, against 42% in the bottom two, and 32% of the tertiary educated against 38% of those with at most primary schooling. The pandemic-era movement runs the other way. Among respondents with a migration intention, 47–48% in the top three wealth quintiles report that the pandemic strengthened their desire to leave, against 37–39% in the bottom two. By education the gradient is monotonic and steep: 34% among those with primary schooling or less, 49% among the secondary educated, and 61% among the tertiary educated, of whom only 4% report a weakened desire.

The two dimensions do not carry equal evidential weight. The education gradient survives the full set of controls: tertiary-educated respondents are 9.7 points more likely to report an intensified intention than those with primary education or less (Table 4, column 4). The wealth gradient is descriptive; conditional on the other covariates, the wealth index does not predict an intention increase. The compositional claim therefore rests on education.

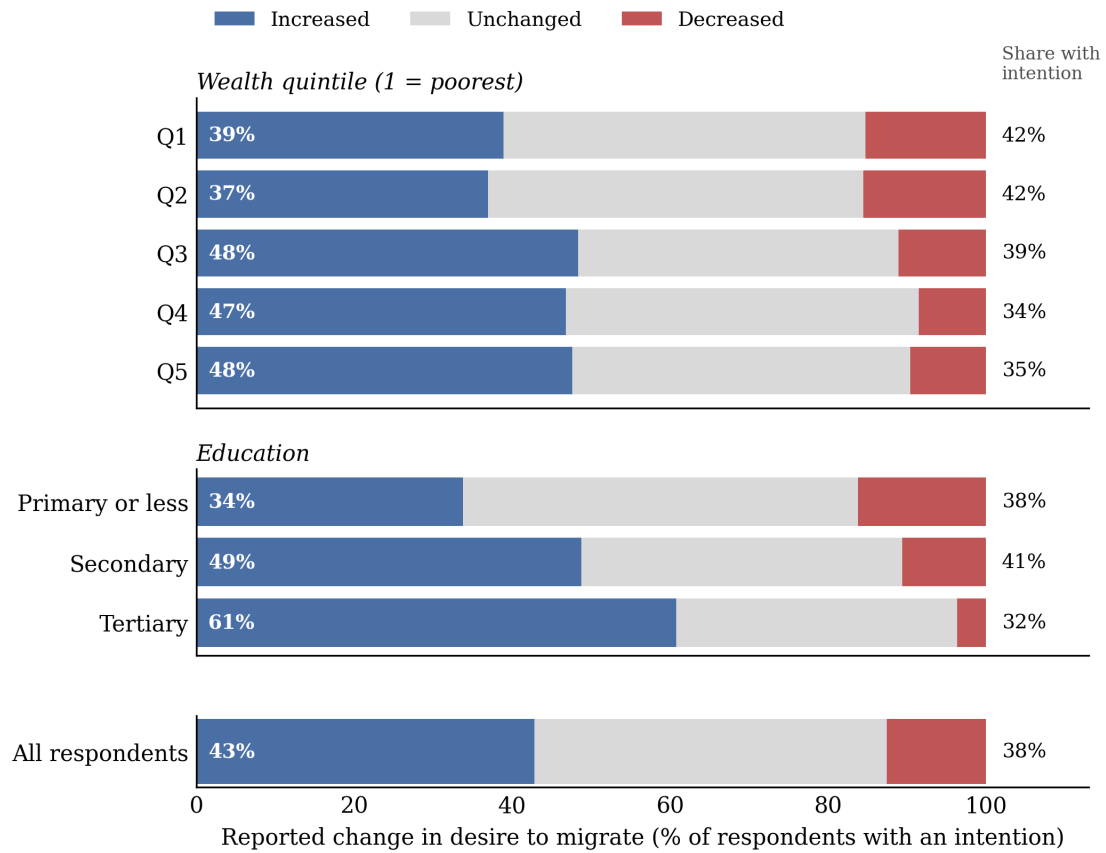


Figure 3: Reported change in the desire to migrate during the pandemic

Notes: Bars show, among respondents expressing a migration intention, the share reporting that the pandemic increased, left unchanged, or decreased their desire to migrate. The right margin reports the share of all respondents in each group expressing a migration intention. Wealth quintiles are computed within country from the principal-component wealth index. *Source:* Authors' calculations from the HFPS Phase 2, wave 2.

The educated and the wealthy have little direct exposure to hunger, so their response is unlikely to be a coping strategy in the usual sense. The reading with most support in the literature is that intentions respond to food insecurity in a respondent's reference group rather than only in their own household. Smith and Wesselbaum (2022) find that food insecurity *inequality* raises migration, arguing that individuals compare themselves with peers, and Nikolova (2023) documents that inequality predicts potential emigration across the Gallup World Poll. On this reading, visible deterioration in the surrounding environment shifts beliefs about the country's trajectory, and those beliefs move intentions even among households insulated from the shock itself.

If sustained, this would matter for the composition of migration rather than only its volume. Better-educated emigration carries familiar consequences for sending countries: shortages in health, education and technical occupations, weaker innovative capacity, and erosion of the social capital held by people who tend to occupy community leadership

roles. Remittances offset the income loss but not the skill loss, so the immediate financial gain and the longer-run developmental cost need not net out. The magnitude is not trivial: a 9.7 point differential on a base of 20% implies tertiary-educated respondents are roughly half again as likely to report an intensified intention as those with primary schooling, in a region where tertiary attainment is concentrated in exactly the occupations whose loss is hardest to replace.

Two cautions apply. This heterogeneity is descriptive, not causally identified. And because the second outcome is conditional on already intending to migrate, part of the apparent increase among wealthier respondents reflects composition within a smaller conditioning set.

5.6. *From intention to realisation*

Intentions are not migration, and the gap between the two is systematic rather than random. Migrating requires cash for travel, documents and settlement, and a willingness to abandon employment and community ties; where wealth constraints bind, aspirations form without the ability to act on them (Carling and Schewel, 2018; Dustmann and Okatenko, 2014). Food insecurity may therefore raise the desire to leave among precisely the households least able to finance departure (Smith and Floro, 2020). The heterogeneity documented above, however, cuts the other way: the group whose intentions moved most, the better educated, is also the group most likely to convert intention into departure. An 11.9 point increase in intentions therefore does not map one-for-one into departures, but neither can it be dismissed as cheap talk: at the aggregate level, changes in stated emigration plans predict changes in realised flows with an elasticity of roughly 0.75 (Tjaden et al., 2019).

We cannot observe realisation in these data. Aggregate movements after the survey window are consistent with the mechanism, encounters at the southern border of the United States rose 27% between 2021 and 2022, with the five countries studied here among the largest contributors (Customs and Border Protection, 2023), but border statistics conflate migration pressure with enforcement, policy and route conditions, and nothing in our analysis identifies the contribution of food insecurity to them. We treat these aggregates as context; the intention results are the paper's evidence.

5.7. *Robustness*

We provide several self-contained robustness checks in detail in the Appendix. Here is the general idea of them: Probit average partial effects for all four specifications of Table 4 are nearly identical to the linear probability estimates (11.0 against 11.9 points for severe food insecurity on the intention level), and re-estimating the main specification excluding one country at a time leaves that coefficient between 10.3 and 14.3 points (Appendix A). Clustering at the department level barely moves the standard errors, and the estimates are robust (mostly) to weighting (Appendix B, Appendix E). The coefficients survive a Romano-Wolf correction for multiple testing (Appendix C), and Oster bounds indicate that unobserved selection would have to be two to three times as strong as selection on the full control set to account for them (Appendix D).

6. Conclusion and policy implications

Using surveys that record food insecurity both currently and retrospectively, we find that severe current food insecurity is associated with an 11.9 percentage point increase in the probability of intending to emigrate, against 4.6 points for pre-pandemic food insecurity, and that only the current measure predicts a pandemic-driven increase in that intention. Classifying households by their food security transition makes the point most sharply: the rise in intentions is concentrated among households the pandemic pushed into food insecurity, not among the chronically deprived. Intentions also rose most among wealthier and better-educated respondents, who are least exposed to hunger directly. The estimates are conditional associations, not causal effects, and the contrast between the two measures is more credible than either level; we read the policy implications in that spirit.

6.1. *Timing, not only volume*

If migration intentions respond to deterioration rather than to accumulated deprivation, the moment at which assistance arrives matters more than its cumulative generosity. Cash transfers reliably raise food consumption in these settings (Bastagli et al., 2019), but a standing programme sized to the chronically poor does not, by construction, reach a household whose food security collapsed six months ago. The margin our results identify is the speed with which a system can expand, not the depth of the benefit it pays (Oxford Policy Management, 2018).

The transition result sharpens this into an operational claim. The households whose intentions moved were food secure before the shock. They therefore held no benefit, appeared in no targeting registry, and in most cases would have failed a means test administered on pre-pandemic data. What binds is not the budget but the capacity to identify and enrol new households within months, the registries, payment rails and eligibility triggers that determine whether a system can expand horizontally when a covariate shock arrives. The emergency transfers improvised across the region in 2020 were, in effect, a test of exactly that capacity, and their coverage varied widely.

A qualification belongs here. This argument presumes intentions translate into departures at a roughly stable rate, which our data cannot verify. The aggregate evidence linking stated plans to realised flows suggests the translation is real (Tjaden et al., 2019), but an intention averted is not the same as a departure averted, and we make no claim about the cost per migration deterred.

6.2. *A mismatch between where assistance goes and whose intentions moved*

The second implication concerns targeting. Intentions moved most among households with little direct exposure to hunger, which points to a response to visible deterioration in the surrounding environment rather than to private consumption alone. External food assistance to the region, by contrast, is directed through humanitarian channels at the areas of chronic deprivation identified before the shock (Husain et al., 2020). Assistance targeted narrowly on food-insecure households will therefore not reach the group whose intentions moved most, and interventions that are visible and broad-based may matter beyond their direct incidence, because what appears to move beliefs is the state of the surrounding environment, not only the state of one's own pantry.

We hold this argument to a lower evidential standard than the main result, and say so. The education gradient survives the full set of controls; the wealth gradient is descriptive. Nor do we observe beliefs directly, so the informational mechanism is an interpretation consistent with the pattern rather than a tested channel. What the pattern does establish is that a food security shock need not act only on the households it makes hungry.

A related observation concerns remittances. Inflows to these countries proved resilient through the pandemic (Knomad, 2022b; Mora-Rivera and Van Gameren, 2021), so the deterioration we study occurred despite the region’s principal private insurance mechanism holding up. Whatever migration pressure the shock generated, it was pressure that existing transfer flows did not absorb.

6.3. *Composition, and a cheap indicator*

If the compositional shift persists, it matters for what emigration costs sending countries rather than only for how much of it there is. The better educated are the likeliest both to form new intentions after a shock and to convert them into departures, since realising an intention requires resources that the poorest lack (Dustmann and Okatenko, 2014; Carling and Schewel, 2018). Losses concentrated in health, education and technical occupations are not replenished by remittance inflows, which offset income but not skill. This argues for monitoring the composition of emigration after shocks, not merely its volume.

It also suggests a low-cost instrument. The migration items we exploit occupied a few minutes of a phone questionnaire, and the phone-survey infrastructure built during the pandemic still exists and has since been redeployed to other acute shocks (Abay et al., 2023). Embedding a short intention module in routine high-frequency surveys would give governments and agencies a leading indicator of migration pressure at negligible marginal cost, one that moves before border statistics do, and that distinguishes a rise driven by newly deprived households from a rise among the persistently poor. Our results suggest those two are different phenomena calling for different responses.

6.4. *Limitations and future work*

The clear priority is data linking intentions to realised migration for the same households. Panel data spanning a shock would allow the current-versus-structural contrast to be estimated with household fixed effects, removing confounders a cross-section cannot, and would replace our recalled pre-pandemic measure with an observed one. Absent that, the direction we would prioritise is measuring the capability margin alongside the aspiration margin (de Haas, 2021): our second outcome captures who *wants* to leave more than before, and says nothing about who can.

Two questions follow directly. Whether the compositional shift persists as the pandemic recedes, or proves a transient response to an exceptional period, is open and consequential. And whether the current-versus-structural asymmetry generalises beyond a pandemic is testable wherever food security deteriorates sharply and is measured twice, price spikes and climate shocks being the obvious cases, and both recurrent (Cattaneo and Peri, 2016; Meierrieks, 2021). If the asymmetry holds there too, the case for building social protection that scales on a trigger rather than on a budget cycle is correspondingly stronger.

CRediT authorship contribution statement

Jorge Soler-Lopez: Conceptualization, Methodology, Software, Formal analysis, Data curation, Writing, review & editing, Visualizations.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics

The analysis uses secondary, de-identified survey data collected and distributed by the World Bank and partners. Informed consent was obtained by the survey implementers at the point of collection. No new data were collected for this study and no ethics approval was required.

Data availability

The High Frequency Phone Survey microdata are publicly available from the World Bank Microdata Library. Census-based municipal migration rates were constructed from national statistical office sources and IPUMS International. Replication code will be made available in a public repository upon acceptance.

Declaration of generative AI in the writing process

During the preparation of this work the author used generative AI tools (Claude Opus 4.8, ChatGPT 4) to assist with language editing and with drafting and checking statistical analysis code. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

570 **Appendix A. Robustness checks**

571 Table Appendix A.1 reports probit average partial effects for the four specifications of Table 4. Table Appendix A.2
572 re-estimates the severe food insecurity specification excluding one country at a time.

Table Appendix A.1: Probit average partial effects				
Dependent Variable	(1)	(2)	(3)	(4)
	Migration intention	Migration intention up since pandemic	Migration intention	Migration intention up since pandemic
Severe FI			0.110*** (0.024)	0.048* (0.019)
Moderate FI	0.088*** (0.015)	0.046*** (0.012)		
FI Pre-Pandemic	0.033* (0.017)	-0.019 (0.014)	0.045** (0.016)	-0.012 (0.013)
Full controls	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes
Observations	5263	5302	5263	5302

Notes: Average partial effects from probit models with the same covariates as Table 4. Delta-method standard errors in parentheses. Significance: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The dependent variable is *migration intention* (columns 1, 3) or *increase in migration intention due to the pandemic* (columns 2, 4).

Table Appendix A.2: Leave-one-country-out estimates, severe food insecurity

Excluded country	Migration intention		Migration intention up since pandemic	
	Severe FI	FI Pre-Pand.	Severe FI	FI Pre-Pand.
Guatemala	0.109*** (0.029)	0.058** (0.020)	0.037 (0.026)	-0.003 (0.017)
	N = 3797		N = 3836	
El Salvador	0.122*** (0.028)	0.046* (0.019)	0.066** (0.025)	-0.022 (0.016)
	N = 4472		N = 4511	
Honduras	0.103** (0.031)	0.025 (0.020)	0.059* (0.028)	-0.020 (0.016)
	N = 4341		N = 4380	
Nicaragua	0.126*** (0.028)	0.053** (0.019)	0.060* (0.026)	-0.014 (0.016)
	N = 4421		N = 4460	
Dominican Rep.	0.143*** (0.035)	0.050* (0.021)	0.051 (0.032)	-0.000 (0.017)
	N = 4021		N = 4021	

Notes: Each row re-estimates columns 3 and 4 of Table 4 excluding the named country. Robust standard errors in parentheses. Significance: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix B. Alternative levels of clustering

The main tables report heteroskedasticity-robust standard errors. Table Appendix B.1 re-estimates the same specifications clustering at the department level, the level at which the fixed effects are defined and at which residual correlation in local labour and food market conditions would operate. There are 103 departments across the five countries, comfortably above the range in which cluster-robust inference becomes unreliable. Point estimates are unchanged by construction; only the standard errors move, and they move very little. The single exception is severe food insecurity on the intention increase, which falls from the 5% to the 10% level.

We also report a more demanding alternative: clustering at the country level. With five clusters the usual cluster-robust asymptotics fail, so we report wild-cluster bootstrap p -values and confidence sets (Cameron et al., 2008). Two points should be read alongside these. First, with five clusters and Rademacher weights there are only $2^5 = 32$ distinct sign vectors, so the bootstrap enumerates the full set and p -values are multiples of $1/32 \approx 0.031$; the resolution is coarse by construction. Second, this is a deliberately conservative exercise, since department fixed effects already absorb all country-level variation, and the five countries were not sampled from a population of countries.

586 The pattern is informative. Every coefficient the paper relies on survives country-level clustering except one:
587 severe food insecurity on the intention increase, whose confidence set covers zero ($p = 0.125$). The transition spec-
588 ification does not share this weakness. Being *newly* food insecure remains strongly associated with an intensified
589 intention ($p < 0.031$, confidence set $[0.061, 0.106]$), and being chronically food insecure remains indistinguishable
590 from never having been food insecure ($p = 0.250$). The specification the paper treats as its sharpest test is also the
591 one that survives the most demanding inference.

Table Appendix B.1: Robust and clustered standard errors

	Migration intention		Intention increased	
	Robust	Cluster	Robust	Cluster
<i>Panel A. Severe food insecurity</i>				
Severe food insecurity	0.119*** (0.027)	0.119*** (0.029)	0.051* (0.024)	0.051+ (0.026)
FI pre-pandemic	0.046** (0.018)	0.046* (0.019)	-0.011 (0.015)	-0.011 (0.015)
<i>Panel B. Moderate food insecurity</i>				
Moderate food insecurity	0.089*** (0.016)	0.089*** (0.018)	0.050*** (0.014)	0.050*** (0.014)
FI pre-pandemic	0.034+ (0.018)	0.034+ (0.018)	-0.020 (0.015)	-0.020 (0.016)
<i>Panel C. Food security transition</i>				
Newly food insecure	0.100*** (0.021)	0.100*** (0.021)	0.080*** (0.019)	0.080*** (0.018)
Chronically food insecure	0.116*** (0.023)	0.116*** (0.022)	0.023 (0.019)	0.023 (0.019)
No longer food insecure	0.054* (0.024)	0.054+ (0.028)	0.009 (0.020)	0.009 (0.018)
Full controls	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes
Clusters	—	103	—	103
Observations	5,264	5,264	5,215	5,215

Notes: Standard errors in parentheses. Significance: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. *Robust* columns are heteroskedasticity-robust; *Cluster* columns cluster at the department (first-level administrative unit). Point estimates are identical across columns. Panels A and B use the specification of Table 4; Panel C uses the transition specification of Table 5, in which the pre-pandemic indicator is subsumed by the transition categories and the omitted group is households food insecure in neither period.

Table Appendix B.2: Wild-cluster bootstrap at the country level

Coefficient	Outcome	p -value	95% confidence set
Severe food insecurity	Migration intention	< 0.031	[0.070, 0.199]
Severe food insecurity	Intention increased	0.125	[−0.090, 0.160]
Moderate food insecurity	Migration intention	< 0.031	[0.038, 0.155]
Moderate food insecurity	Intention increased	< 0.031	[0.019, 0.079]
Newly food insecure	Migration intention	< 0.031	[0.036, 0.171]
Newly food insecure	Intention increased	< 0.031	[0.061, 0.106]
Chronically food insecure	Migration intention	< 0.031	[0.065, 0.206]
Chronically food insecure	Intention increased	0.250	[−0.036, 0.069]

Notes: Wild-cluster bootstrap- t with the null imposed, clustering on country ($G = 5$), Rademacher weights, full enumeration of the $2^5 = 32$ sign vectors. The finest attainable p -value is therefore $1/32 \approx 0.031$, and entries reported as < 0.031 are cases in which no enumerated draw exceeded the sample statistic. Specifications as in Table Appendix B.1.

Appendix C. Multiple hypothesis testing

We report four principal specifications, two food insecurity measures crossed with two outcomes, plus a transition variant. Testing several hypotheses on one sample inflates the chance that at least one crosses conventional significance by luck. A Bonferroni correction would address this but is badly suited here: our tests use overlapping samples and share a control set, so they are strongly positively correlated and Bonferroni is correspondingly conservative.

We therefore report Romano–Wolf stepdown adjusted p -values (Romano and Wolf, 2005), computed with the procedure of Clarke et al. (2020) using 2,500 bootstrap resamples clustered at the department level. The method bootstraps the family of tests jointly, so the correlation between them is estimated rather than assumed, and it proceeds stepwise: the most significant coefficient is compared with the bootstrap distribution of the maximum test statistic across the family, and if rejected is removed before the procedure repeats on the remainder. This controls the family-wise error rate, the probability of even one false rejection anywhere in the family, while retaining more power than a single-step correction.

Table Appendix C.1 reports two families. Family A contains the eight coefficients on which the paper’s contrast rests: current and pre-pandemic food insecurity, in each of the four specifications of Table 4. Family B contains the six transition coefficients of Table 5.

The contrast survives correction. In Family A, current food insecurity remains significant on both outcomes, and pre-pandemic food insecurity remains significant on the intention level while remaining far from significance on the intention increase. In Family B, being newly food insecure is significant on both outcomes at $p = 0.0004$, while chronic food insecurity is insignificant on the intention increase. The asymmetry the paper documents is not an artefact of having estimated several specifications.

Adjusted p -values below the unadjusted ones, which occur for two coefficients, are not an error. The bootstrap resample p -value is a separate and, in this setting, more accurate approximation than the analytic clustered p -value; the Romano–Wolf adjustment is applied to the former, and can lie below the latter when the analytic approximation is conservative.

Table Appendix C.1: Romano–Wolf stepdown adjusted p -values

Coefficient	Outcome	Unadjusted	Resample	Romano–Wolf
<i>Family A. Current and pre-pandemic food insecurity (8 hypotheses)</i>				
Severe food insecurity	Migration intention	0.0001	0.0004	0.0004
FI pre-pandemic	Migration intention	0.0146	0.0004	0.0048
Severe food insecurity	Intention increased	0.0527	0.0100	0.0256
FI pre-pandemic	Intention increased	0.4670	0.3403	0.3403
Moderate food insecurity	Migration intention	0.0000	0.0004	0.0004
FI pre-pandemic	Migration intention	0.0614	0.0088	0.0256
Moderate food insecurity	Intention increased	0.0004	0.0004	0.0004
FI pre-pandemic	Intention increased	0.2195	0.1000	0.1220
<i>Family B. Food security transition (6 hypotheses)</i>				
Newly food insecure	Migration intention	0.0000	0.0004	0.0004
Chronically food insecure	Migration intention	0.0000	0.0004	0.0004
No longer food insecure	Migration intention	0.0571	0.0112	0.0300
Newly food insecure	Intention increased	0.0000	0.0004	0.0004
Chronically food insecure	Intention increased	0.2358	0.1100	0.2007
No longer food insecure	Intention increased	0.6170	0.4770	0.4770

Notes: 2,500 bootstrap resamples, clustered at the department level, seed 20260806. *Unadjusted* is the analytic p -value from the clustered regression; *Resample* is the bootstrap p -value for the single hypothesis; *Romano–Wolf* is the stepdown adjusted p -value controlling the familywise error rate within the family shown. Families are corrected separately. Specifications as in Tables 4 and 5.

Appendix D. Selection on unobservables

The estimates are conditional associations, and the obvious threat is a confounder we do not observe. Oster (2019) provides a way to discipline that concern using information the estimation already generates: how much the coefficient of interest moved when observed controls were added, and how much additional variance those controls explained. If a coefficient barely moves while the controls explain a great deal, then unobservables of comparable character would have to be implausibly powerful to drive it to zero.

Two quantities follow. The first, δ , is the strength of selection on unobservables relative to selection on observables that would be required to reduce the coefficient to zero; $\delta = 1$ means unobservables would have to matter exactly as much as the full control set, and values above one are conventionally taken as reassuring. The second, β^* , is the

bias-adjusted coefficient under $\delta = 1$: if the identified set between the controlled estimate and β^* excludes zero, the result survives the assumption that unobservables matter as much as observables.

Both depend on $R_{\max}$, the R^2 a regression including all relevant controls would attain. It is not observable. Oster's convention, calibrated against replications of randomised results, is $R_{\max} = \min(1.3\tilde{R}^2, 1)$. Because that choice does real work, we report it alongside two more demanding alternatives rather than presenting a single number.

Table Appendix D.1 reports the results. At Oster's conventional bound, every coefficient returns δ between 1.9 and 2.9: unobserved selection would need to be roughly two to three times as important as the entire observed control set, which includes wealth, education, income loss, government assistance, remittance receipt and department fixed effects. At $R_{\max} = 1.5\tilde{R}^2$ all five remain above one. At $R_{\max} = 2\tilde{R}^2$ all fall below one, though that bound requires a hypothetical full model explaining roughly 27 to 31 per cent of the variance in a binary outcome against the 13 to 15 per cent achieved here, which we regard as demanding for survey data of this kind. The coefficient the paper leans on most—being newly food insecure, on the intention increase—is also the most stable, with $\delta = 2.92$ and an identified set of $[0.057, 0.080]$ excluding zero.

Two limitations should be stated rather than left implicit. First, the procedure bounds a single coefficient, whereas the paper's claim is a contrast between two. Bounding β_1 does not bound $\beta_1 - \beta_2$, so Table Appendix D.1 should be read as a check on the individual estimates and not as a defence of the contrast, whose credibility rests on the argument in Section 4 instead. Second, proportional selection is an assumption about the data-generating process, not a testable property, and it is a stronger assumption here than in a setting with an assigned treatment.

Table Appendix D.1: Oster bounds on selection on unobservables

Coefficient	Outcome	Inputs		δ for $\beta = 0$			β^*
		$\tilde{\beta}$	$\tilde{R}^2$	$1.3\tilde{R}^2$	$1.5\tilde{R}^2$	$2\tilde{R}^2$	at $\delta = 1$
Severe FI	Intention	0.119	0.151	2.77	1.71	0.88	0.081
Severe FI	Intention increased	0.051	0.134	2.75	1.67	0.84	0.035
Moderate FI	Intention	0.089	0.153	1.90	1.19	0.61	0.051
Moderate FI	Intention increased	0.050	0.136	1.93	1.18	0.60	0.029
Newly FI	Intention increased	0.080	0.137	2.92	—	—	0.057

Notes: $\tilde{\beta}$ and $\tilde{R}^2$ are the coefficient and R^2 from the controlled regression, i.e. the specification of Table 4 for the first four rows and of Table 5 for the last. Department fixed effects are treated as design rather than as informative controls and are included in both the uncontrolled and the controlled regression. δ is the relative degree of selection on unobservables that would drive the coefficient to zero, at the three values of $R_{\max}$ shown. β^* is the bias-adjusted coefficient setting $\delta = 1$ and $R_{\max} = 1.3\tilde{R}^2$. Computed with `psacalc`.

Appendix E. Weighted estimates

The estimates in the main text are unweighted, for the reasons set out in Section 4. Table Appendix E.1 re-estimates every specification using the household sampling weights supplied with the survey, with heteroskedasticity-robust standard errors.

The coefficients of interest are unchanged in substance. Severe food insecurity is associated with an 11.2 point increase in the probability of intending to emigrate, against 11.9 points unweighted, and moderate food insecurity with 9.4 points against 8.9. On the intention increase, the weighted estimates are slightly larger and slightly more precise than the unweighted ones: severe food insecurity moves from 5.1 to 6.0 points and regains significance at the 5% level, which it loses under department clustering. The transition estimates are nearly identical, with newly food insecure households 7.7 points more likely to report an intensified intention against 8.0 points unweighted, and chronically food insecure households indistinguishable from the never food insecure in both.

One coefficient changes in a way worth stating. Pre-pandemic food insecurity, which is statistically indistinguishable from zero on the intention increase in the unweighted specifications (-0.011 and -0.020), becomes significantly negative once weights are applied (-0.032 and -0.040 , both $p < 0.05$). Taken at face value this strengthens rather than qualifies the paper's argument: households that were already food insecure before the pandemic were, if anything, *less* likely to report that the pandemic increased their desire to leave. We do not lean on this. The weights are constructed to recover population means and are not designed for conditional estimation, so a coefficient that is null unweighted and small but significant weighted is better read as bounded near zero than as evidence of a negative relationship. What matters for the paper is that the pre-pandemic measure never behaves like the current measure on this outcome, and that is true under both weighting schemes.

Table Appendix E.1: Unweighted and weighted estimates

	Migration intention		Intention increased	
	Unweighted	Weighted	Unweighted	Weighted
<i>Panel A. Severe food insecurity</i>				
Severe food insecurity	0.119*** (0.027)	0.112*** (0.031)	0.051* (0.024)	0.060* (0.028)
FI pre-pandemic	0.046** (0.018)	0.037+ (0.021)	-0.011 (0.015)	-0.032* (0.016)
<i>Panel B. Moderate food insecurity</i>				
Moderate food insecurity	0.089*** (0.016)	0.094*** (0.019)	0.050*** (0.014)	0.051** (0.016)
FI pre-pandemic	0.034+ (0.018)	0.023 (0.021)	-0.020 (0.015)	-0.040* (0.017)
<i>Panel C. Food security transition</i>				
Newly food insecure	0.100*** (0.021)	0.102*** (0.025)	0.080*** (0.019)	0.077*** (0.022)
Chronically food insecure	0.116*** (0.023)	0.121*** (0.026)	0.023 (0.019)	0.007 (0.021)
No longer food insecure	0.054* (0.024)	0.024 (0.028)	0.009 (0.020)	-0.018 (0.021)
Full controls	Yes	Yes	Yes	Yes
Department fixed effects	Yes	Yes	Yes	Yes
Observations	5,264	5,264	5,215	5,215

Notes: Heteroskedasticity-robust standard errors in parentheses. Significance: + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Weighted columns use the household sampling weight supplied with the HFPS Phase 2 wave 2 data. Panels A and B follow the specification of Table 4; Panel C follows Table 5, where the omitted category is households food insecure in neither period.

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